

Target Information and Profile System (TIPS) Prototype

Usability Evaluation

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Abstract

This document discusses the results of a usability evaluation to obtain feedback from Small Business Group (SBG) sales representatives on Target Information and Profile System Prototype in Phoenix, Minneapolis and Seattle. Questionnaires and individual interviews assessed the usability and likability of TIPS panels and the profile items and their categories. Comments were also solicited on design and navigational issues.

Tables of Contents

1.0 Executive Summary.....	1
1.1 Goal of the Evaluation.....	1
1.2 Method	1
1.3 Results.....	1
1.4 Recommendations.....	3
2.0 Introduction.....	5
2.1 Purpose and Scope.....	5
2.2 Author	5
2.3 System and Interface Description	5
2.4 Evaluation constraints	5
3.0 Evaluation Methods.....	6
3.1 Questionnaires.....	6
3.2 Individual Interviews	7
3.3 Subjects	7
3.4 Procedure	7
4.0 Results and Analysis.....	7
4.1 Questionnaire Results.....	7
4.3 Overall Usability of TIPS	12
4.4 Usability and Likability of Individual Panels	13
4.5 Individual Interview Comments.....	14
4.5.1.1 Profile Items and Categories.....	14
4.5.1.2 Business Info.....	17
4.5.1.3 Concerns	18
4.5.1.4 Adding.....	20
4.5.1.5 Easy Calling	20
4.5.1.6 In / Out Usage.....	21
4.5.1.7 Usage.....	22
4.5.1.8 Identify Calls	23
4.5.1.9 Control Calls.....	24
4.5.1.10 Ans Methods.....	24
4.5.1.11 Out of Office	25
4.5.1.12 Equipment.....	25
4.5.1.13 Image.....	26
4.5.1.14 Visibility	27
4.5.1.15 Advertising	27
4.5.1.16 Client Base	27
4.5.1.17 General Comments and Suggestions.....	28
4.6 Summary & Recommendation Panels Comments.....	28
4.7 Visual Organization and User Interface Elements Comments.....	29
4.8 Task Compatibility and Work Flow	31
4.9 Overall System Comments.....	32
4.10 Overall System Value	33
4.11 Other Suggestions from CCOE Observers	34
5.0 General Recommendations and Conclusions	34

Appendix A Interview Questions.....	35
Appendix B Usability Inventories.....	40
Appendix C TIPS Panels and Windows	50

1.0 Executive Summary

1.1 Goal of the Evaluation

The goal of this usability evaluation was to obtain feedback from Small Business Group (SBG) sales representatives on the Target Information and Profile System (TIPS) Prototype. TIPS is the sales assistant module of SSNQ and uses a graphical user interface designed to support consultative sales activities of the sales representatives (sales reps) in the business service centers. The sales reps can collect, view and update customer site profile information through the profile panels in the TIPS Manager window. The underlying knowledge-based engine will provide a set of possible product recommendations appropriate to the needs of the customers from their profile information. This evaluation focused on the usability of the profile items and categories as well as the profile items panels, the summary panel (under TIPS Manager window) and the recommendation window.

1.2 Method

Eighteen SBG sales representatives from the Phoenix, Minneapolis and Seattle Business Offices responded to written usability questionnaires and participated in individual interviews. The questionnaires evaluated the overall usability of TIPS and the usability, likability, and other general usability issues of each individual window. Comments were solicited on individual panels, specific profile items and categories, window design and navigational issues and the overall interface during the interviews.

1.3 Results

Overall, the results from the usability testing were positive. The sales representatives (sales reps) rated the majority of the windows and profile items as easy to use. Comments on the profile items windows revealed that some of the sales reps question the amount of information that they gather and the resulting impact on customer contact time.

The ratings from the general usability questionnaire were generally favorable. The sales reps feel that the profile panels will prompt them to ask more consultative questions and the summary panel will help them to establish rapport quickly with the customers in subsequent contacts. Most of them like the product benefits offered by the recommendation panel. When they were asked whether they will use TIPS to support their consultative sales contact with their customers (in questionnaires), 14 out of 18 (78%) reps responded that they would. One had no opinion (5%) and the other three (17%) had negative responses. Fourteen of the respondents (78%) thought that TIPS would add more value to consultative sales. Two had (11%) no opinion and the other two (11%) had negative responses.

In the interviews, some of the sales reps expressed concerns on the adjustments of work style and the time required to complete TIPS during the customer contact.

Comments and suggestions recorded in the individual interviews are presented. Suggestions for improvements and negative comments are grouped, tabulated and graphed. These suggestions and negative comments provide valuable information and feedback so that continuous improvement of the system can be implemented. Out of the 82 profile items, 11 items (13%) received 4 or more negative comments and suggestions. Some of the positive comments are also presented. The positive comments showed that TIPS has been developed in the right direction. In addition, recommendations are given for each action item identified in the usability session.

The following chart summarized and compared the results across the 3 cities on the following overall usability questions of TIPS:

- The steps of TIPS are easy to follow
- TIPS is compatible with the flow of consultative sales
- prospective use of TIPS in consultative sales
- TIPS adds more value to consultative sales

The overall results showed that the reps interviewed were generally positive about the system. A seven-point scale is used to represent the extent of agreement to the questions with 7 denoting "strongly agree" and 1 denoting "strongly disagree".

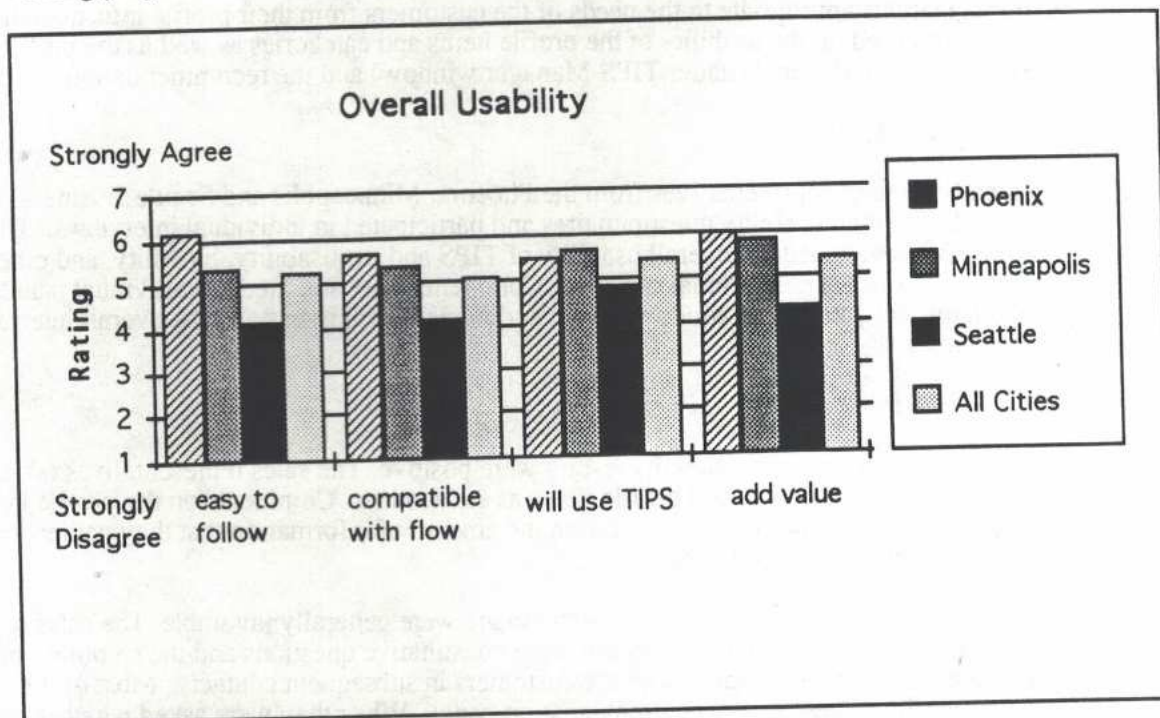


Fig 1. overall usability of TIPS

The charts on the next page summarized the results of the likability and usability of individual panels or windows of TIPS:

- profile panel 1 - business information and concerns
- profile panel 2 - line use
- profile panel 3 - equipment and advertising
- summary panel - summary of the profile items appropriate to customer site
- recommendation window - sets of products fulfilling customer needs, product benefits and needs/profile relationship

Again, the results indicated that the sales reps were positive about individual panels and windows.

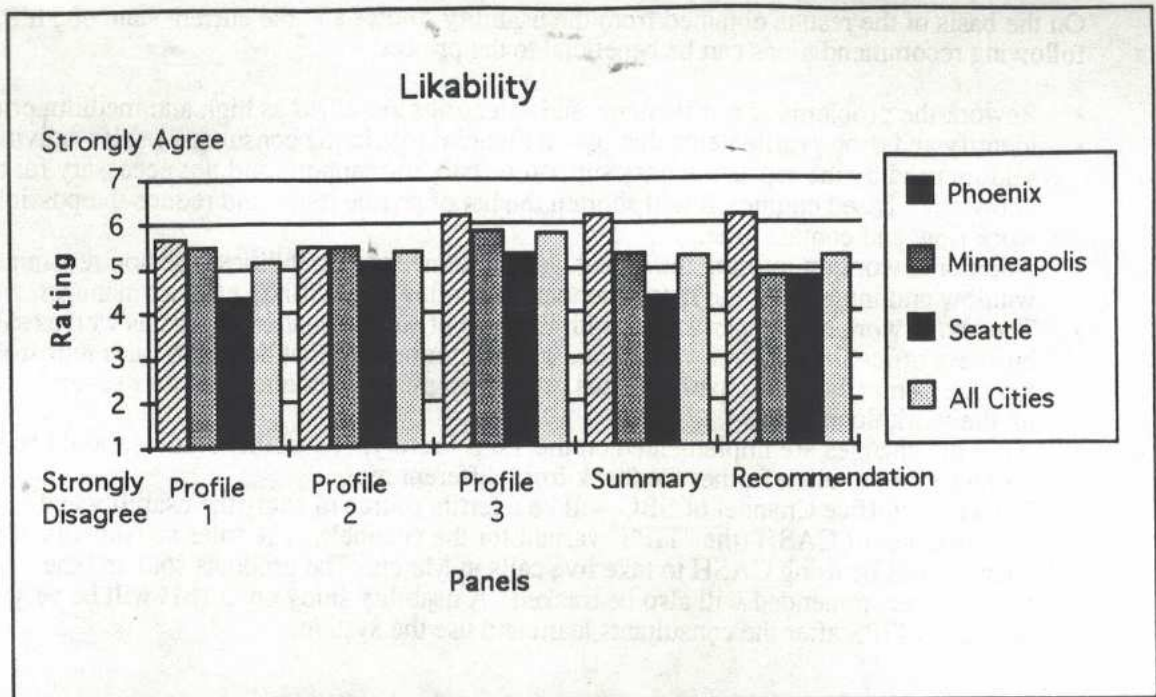


Fig 2. Likability of individual panels of TIPS

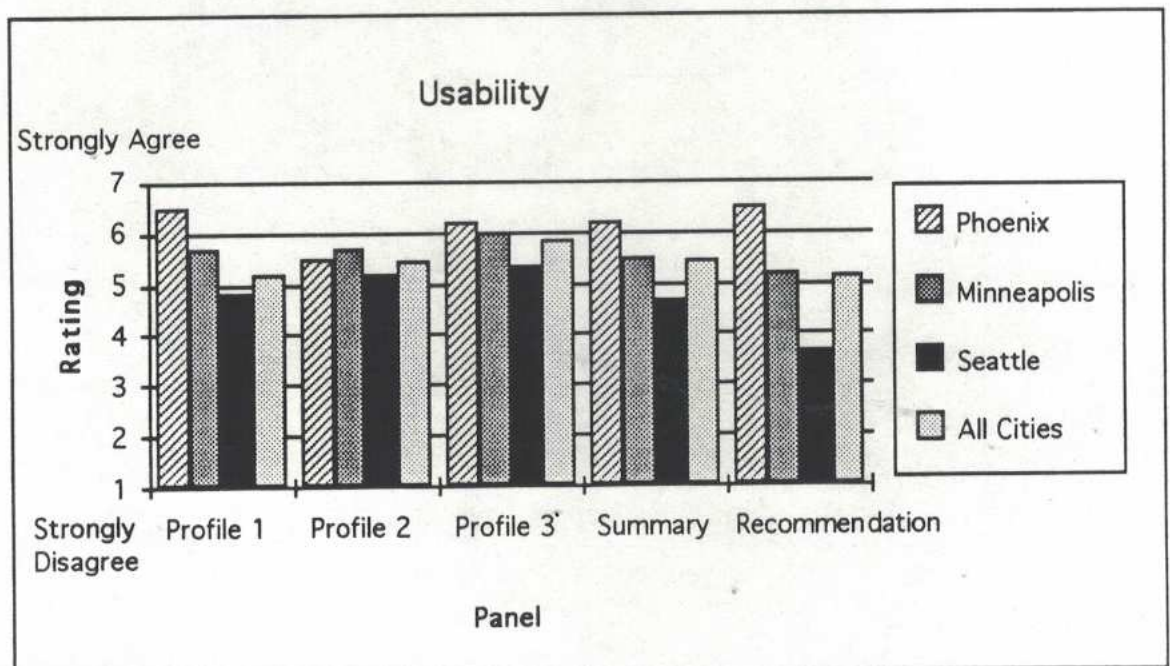


Fig 3. Usability of individual panels of TIPS

More detailed results of the study were presented after the executive summary.

1.4 Recommendations

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On the basis of the results obtained from the usability studies and the current state of TIPS, the following recommendations can be beneficial to the project:

- Rework the problematic profile items and categories identified as high and medium priority.
- Identify and drop profile items that play a minimal role in the consultative sales activities (e.g., seldom used by the rep as memory support or building rapport) and not necessary for the knowledge-based engine. It will shorten the list of profile items and reduce the possible impact on work flow and contact time.
- Additional work on refining the names (labels) of needs (capabilities) used on recommendation window and integrating the recommendation window with SSNQ product manager.
- The above work can be performed with the help of subject matter experts as well as sales reps from business office for informal validation. It is recommended that at least 6 sales reps from Denver Service Center can participate in focus group design and validation sessions several times to iterate on the work done above.
- After the changes are implemented on the TIPS Prototype, usability studies should be conducted in various cities to elicit further feedback from different reps.
- The Home Office Channel of SBG will be a fertile source to study the usability and effectiveness of CASH (the "TIPS" variant for the channel). The sales consultants of the channel will be using CASH to take live calls in March. The products sold and the products recommended will also be tracked. A usability study on CASH will be very helpful to TIPS after the consultants learn and use the system.

2.0 Introduction

2.1 Purpose and Scope

This report presents usability test results for the Target Information and Profile System (TIPS) Prototype. The goal was to gather feedback from users on profile items and categories, summary and recommendation, specific window features and issues of screen navigation. The evaluations were conducted with Small Business Service Representatives from the Phoenix, Minneapolis and Seattle Business Service Centers. The following panels were included:

TIPS Windows / Panels	Description of Intended Functionality
TIPS Manager	Top-level TIPS window. Allow the rep to change views to three different profile items panels and summary and launch the recommendation window.
TIPS Manager with Customer Site Profile 1	Allow the rep to collect or view profile items related to business information, layout, concerns, adding services and easy calling features of the customer site.
TIPS Manager with Customer Site Profile 2	Allow the rep to collect or view profile items related to in/out usage, usage, identify calls, control calls, answering methods and out of office needs.
TIPS Manager with Customer Site Profile 3	Allow the rep to collect or view profile items related to equipment, image, visibility, advertising and client base.
TIPS Manager with Summary	Allow the rep to have a bird's eye view of the profile items clicked in current or previous session.
Recommendation	Allow the rep to utilize sets of recommended products appropriate to customer needs for presenting as well as product benefits and needs/profile relationship to overcome objections.

Table 1. TIPS Windows/Panels and intended functions

2.2 Author

The author of this document is Yat-Sang Hung, Member of Technical Staff, Standard Services and Human Factors, U S West Technologies. The author is grateful to the feedback and suggestions of the following persons in the process: Judy Warner, Mike Henry, Cathe Valdez, Chris Plott, Gail Fontenelle, John Bierschwale, Amy Geoffroy, Fred Bak and Nikki Brower.

2.3 System and Interface Description

The TIPS system is described in the *TIPS User Requirements* and the *TIPS User Interface Specification*. The intended functionality allows the sales representative to collect and view customer profile and to utilize the product recommendations, product benefits and needs/profile relationship provided by the underlying knowledge base of the system to conduct consultative sales more effectively.

2.4 Evaluation constraints

The TIPS Prototype was implemented with a simple scenario. There were no functionalities of the TIPS Prototype except simple behavior of the interface objects such as the toggle buttons and option menus. The intended functionalities were demonstrated through a built-in scenario. Thus, the sales

reps could not actually use the prototype to perform sales consultation. In the usability sessions, they expressed the desire to have more hands-on experience with the system.

Eighteen sales reps from 3 cities (Phoenix, Minneapolis, Seattle) across three regions participated in the studies. Subject selection was determined by each individual business office according to time, availability, scheduling and other issues. Since a random selection of subjects was not possible, the results obtained should be interpreted cautiously. Trends and differences among the cities and regions will be investigated with the understanding that rigorous statistical generalizations are impossible given the constraints of the study.

This evaluation was intended to obtain feedback from the sales reps concerning issues of profile items and categories understandability as well as navigation and presentation of specific window features. This type of evaluation must be viewed as one phase in an ongoing program of user interface development and iterative design. Gathering user input at every phase of development ensures a desirable degree of validity and user acceptance.

3.0 Evaluation Methods

This evaluation employed two methods of assessing usability: 1) questionnaires 2) individual interview. The questionnaires provided the users with the opportunity to rate the usability and likability of the system at both an overall and an individual window level. These inventories were administered to all the subjects of the three cities. The individual interviews provided a vehicle through which the users could give subjective open-ended and closed-ended feedback regarding the profile items and categories, the design of the individual windows and some specific issues of the system. Because of the limited time availability for each subject, an iterative design was adopted to maximize the amount and depth of feedback collected at different cities. The questions for individual interviews were iterated throughout the process of the study from tentative results obtained from previous sessions to focus on identified issues.

In the first iteration with the sales reps in Phoenix, an extensive questionnaire on the usability of each category and each profile item (including the understandability of the labels and descriptions and the appropriateness of classification) was administered to define scope of the interview questions used in later iterations (Minneapolis and Seattle). The results from the Phoenix study served to focus attention on problematic items and categories identified. A set of interview questions pinpointing the identified issues was then designed for use in Minneapolis.

The second iterative study took place in Minneapolis where the sales reps were asked in interviews to identify profile items and categories that did not appear self-explanatory. Specific responses and suggestions on problematic items and categories identified in Phoenix were then elicited.

In the third iterative study (Seattle), results and suggestions from the second iterations were integrated into the interview questions to explore those issues in more depth. The questions used in the interviews were included in the appendix.

3.1 Questionnaires

The questionnaires consisted of the following usability inventories to elicit feedback on each individual panel or window. They were administered to all the subjects in the three cities:

- customer site profile panel 1 - business information and concerns
- customer site profile panel 2 - line use
- customer site profile panel 3 - equipment and advertising

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- customer site profile summary panel
- recommendation window
- overall system usability

The first three questionnaires consisted of a 10 item or 11 item likert scale designed to assess various usability aspects of the profile panels. The other two contained similar questions on the summary and recommendation panels. The final inventory consisted of overall usability questions of TIPS. The rating scale used a 7-point likert scale with a score of 7 representing "strongly agree" and a score of 1 representing "strongly disagree". Questions were worded both negatively and positively to avoid rating biases.

3.2 Individual Interviews

The sales reps were interviewed on a one-on-one basis. These interviews were targeted towards eliciting comments about profile items and profile categories, specific window features and issues of window navigation. General comments about the profile items and the windows, if offered by the sales reps, were recorded as well.

3.3 Subjects

The average experience of the sales reps in three different cities was tabulated in the following:

Cities	Years with U S WEST	Years as SBG Rep
Phoenix	16.25	11.9
Minneapolis	18.83	5.41
Seattle	14.5	8.58

Table 2. Average years of experience of sales reps

3.4 Procedure

The usability study was conducted in Phoenix BSC from Jan 12-13, 1994, in Minneapolis BSC from Jan 19-20, 1994 and in Seattle BSC from Jan 26-27, 1994. Six sales representatives from each city participated in the study. A conceptual walkthrough of the system was conducted by the SME to demonstrate the use of TIPS and SSNQ.

Each sales rep was interviewed and was then asked to complete the questionnaires. The prototype system was shown before them all the time and they were free to explore the system when they were completing the questionnaires.

The raw interview data were transcribed and unitized on index cards. They were then sorted into natural categories, classified and analyzed. The data of the usability inventories were also collected and compiled.

4.0 Results and Analysis

The results of the evaluation are presented as follows: Section 4.1 to 4.4 present the results from the usability inventories; Section 4.5 to 4.10 summarize the comments gathered during the individual interviews.

4.1 Questionnaire Results

The following tables and graphs summarized the different areas of coverage of the questions for the profile panels and the overall average ratings of the three cities. Ratings of specific cities are included in the appendix. The numbers with asterisks indicated significant differences of ratings were found across the cities (alpha = 0.05, 1-way ANOVA).

Questions	Profile Panel 1	Profile Panel 2	Profile Panel 3	Purpose
categories & profile items are easy to read	5.67	5.56	5.89 *	visual organization
categories & profile items organized logically	5.28	5.56	5.61*	semantic organization
different shades help to locate appropriate categories	5.11	4.83	4.94	use of color
facilitates flow of consultative sales	4.50	5.00	5.00	task and work flow
prompt to ask more questions	5.50	5.33	5.28	task enhancement

Table 3. Usability questions and ratings of profile panels

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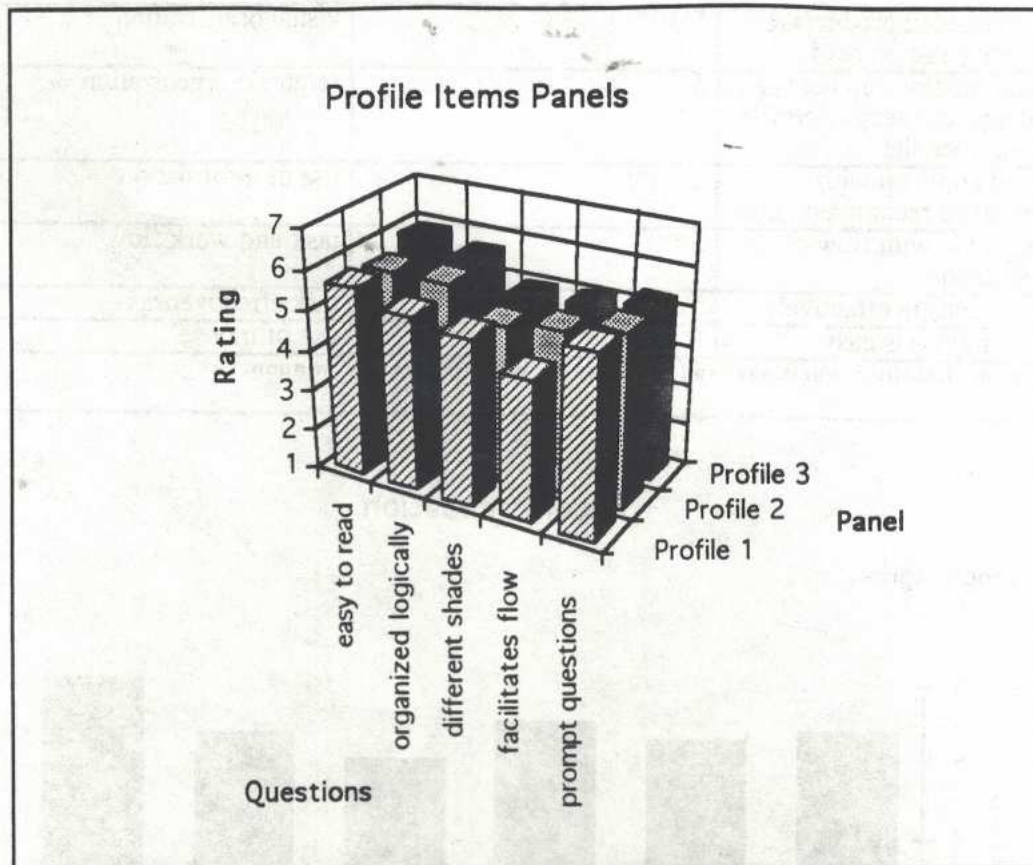


Fig 4. Usability questions and ratings of profile panels

The above figure shows the comparisons of usability indices (ratings) across the three profile panels. The questions on the use of different shades and the facilitation of consultative sales received relatively low ratings. In the interviews with the sales reps, it was found that they like colors of stronger contrast to differentiate different categories. In the facilitation of consultative sales, some reps remarked that the system is not compatible with their current work flow and major adjustments may have to be made. See the Interview Section of this report.

The following table summarizes the results of the summary panel:

Questions	Summary	Purpose
categories & profile items easy to read	5.61	visual organization
bird's eye view important	5.61	global information
facilitates consultative flow	5.5	task and work flow

Table 4. Usability questions and ratings of summary panel.

The overall rating of this panel was positive. The results, together with the responses from the interviews showed that the reps find this panel useful and handy to establish rapport with customers.

The following table summarizes the results of the recommendation window:

Questions	Recommendation	Purpose
-----------	----------------	---------

recommended products & functions easy to read	5.39*	visual organization
logical relationship between products and needs / profile of customer site	5.28	semantic organization
easy to come up with appropriate recommendation	5.50*	use of information
compatible with flow of consultation	5.00	task and work flow
help to consult effectively	5.33*	task effectiveness
use of mouse is easy	6.05	use of mouse

Table 5. Usability questions and ratings of recommendation window.

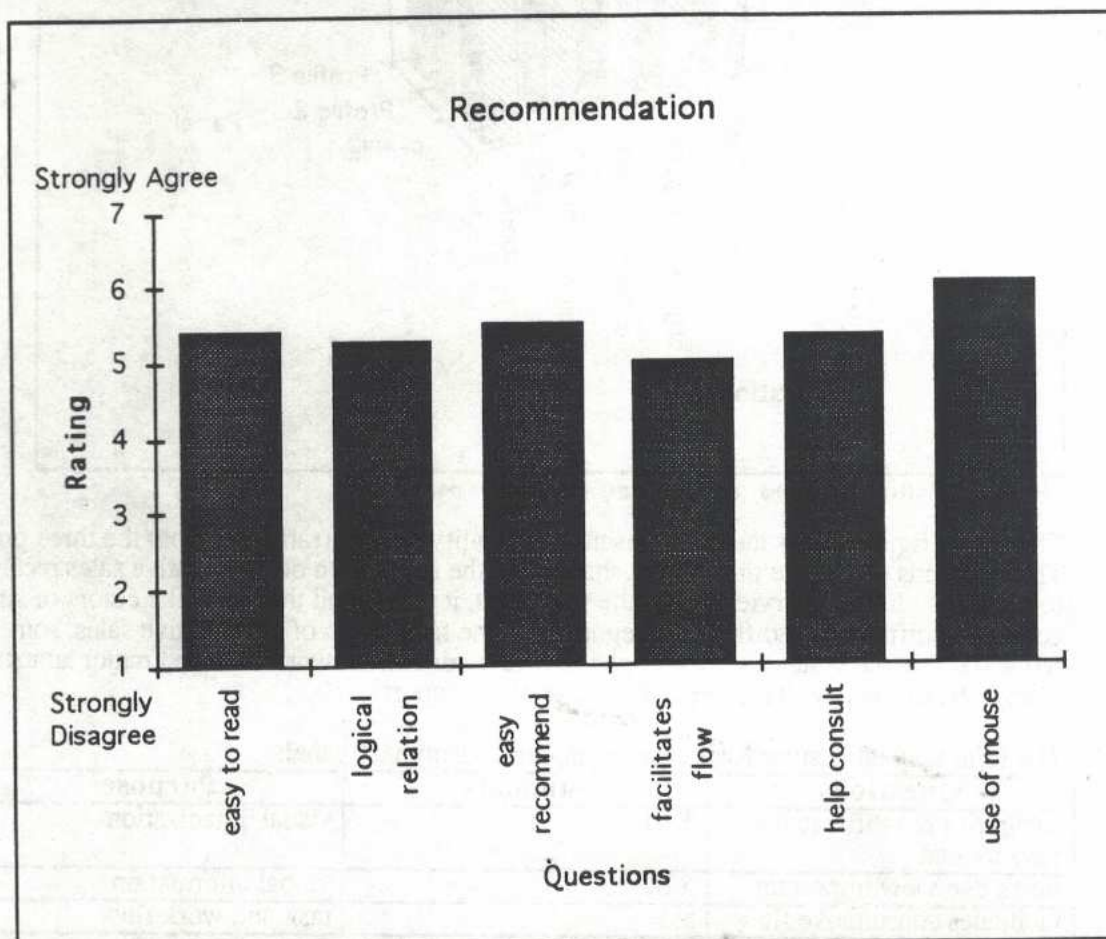


Fig 5. Usability questions and ratings of recommendation window

The overall results of the recommendation panel were positive. The use of mouse was hailed by some reps as a partial solution to the carpal-tunnel syndromes common to the prolonged use of keyboards.

In general, sales reps in Phoenix tended to score higher than the other 2 cities and sales reps in Seattle tended to score lower. This trend was consistent with the results obtained in the usability testing results of SSNQ.

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The following table summarizes the results of the icons (the question used is whether the icon makes sense):

Icons	Rating
Business Info	4.22
Concerns	4.52
Line Use	4.78
Equipment	4.89
Advertising	6.06
Summary	4.06
Recommendation	5.28

Table 6. Usability rating of icons

Thirteen out of 18 sales reps also preferred to have 1-1 correspondence between icons and profile panels. Results from various sources suggested that some of icons need to be reworked and some of those need to be dropped. Specifically, the icons for Business Info and Concerns can be combined as well as the icons for Equipment and Advertising.

The negatively worded questions represented the same trend in the data (* significant at $\alpha = 0.05$, one-way ANOVA). The following summarized the results of negatively worded questions (1 represents strong disagreement with the question):

Questions	Profile 1	Profile 2	Profile 3	Purpose
difficult to find the required profile item or category	3.06	2.29	2.56	visual organization
panel is ill-organized	2.12	2.17	1.94	semantic organization

Table 7. Usability questions and ratings of profile panels (negatively worded questions)

Questions	Summary	Purpose
difficult to find the required profile item or category	2.39	visual organization

Table 8. Usability questions and ratings of summary panel (negatively worded question)

Questions	Recommendation	Purpose
difficult to understand and use layout	3.17*	information organization
prevent conducting consultative sales effectively	2.72	work flow

Table 9. Usability questions and ratings of recommendation window (negatively worded questions)

The questionnaires results obtained were consistent. More work needs to be done on some of the categories and profile items. The presentation of the recommendation window also needs to be fine-tuned.

4.3 Overall Usability of TIPS

The following table summarized the results of the questions directed to explore the overall usability of the system. The ratings of each city were shown in the graph together with the average for all three cities (* significant at alpha = 0.05, one-way ANOVA):

Questions	Overall	Purpose
easily follow the steps of TIPS in customer contact	5.22*	work flow
natural organization of panels and compatibility of flow	5.06	work flow
will use TIPS to support consultative sales	5.33	prospective usage of TIPS
adds more value to consultative sales	5.38*	utility of TIPS

Table 10. Overall usability questions and ratings

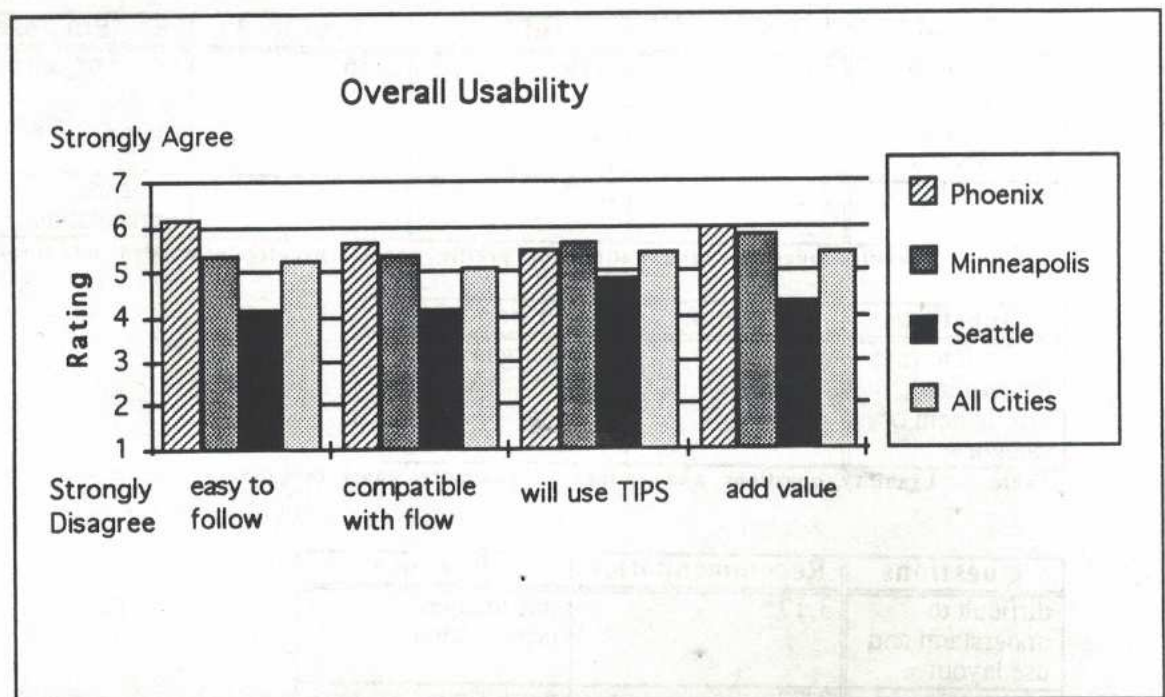


Fig 6. Overall usability questions and ratings of TIPS

Again, sales reps in Seattle tended to score lower on the questions. More in-depth input and suggestions from the sales reps in Seattle (or the Western Region) on the design of the system may be needed.

4.4 Usability and Likability of Individual Panels

The sales representatives were asked to rate the usability and likability of the five panels (3 customer profile information, a summary and a recommendation panel). The likability question asked: "How much do you like this screen?" A seven point rating scale was used with 1 indicating strongly dislike and 7 indicating strongly like. The average scores (all the subjects in all cities) of all the panels were between 5.16 and 5.78 on the 7-point scale.

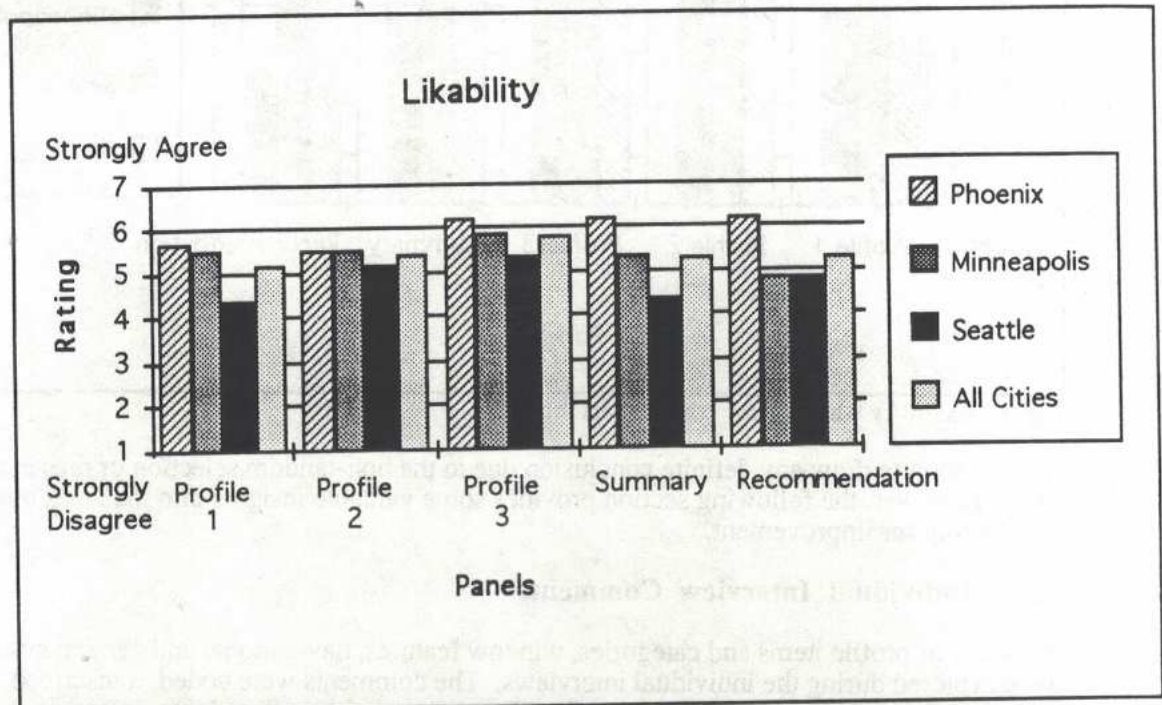


Fig 7. Likability ratings of panels of TIPS

The usability question statement read, "I think this screen would be easy to use." Again, the average scores (all the subjects in all cities) of all the panels were above 5.00 (ranging from 5.11 to 5.83) on the 7-point scale.

When the scores were compared across cities, the average of the sales reps of Seattle had a lower ratings than the other two cities, though the results were only statistically significant for the first profile panel and the recommendation panel.

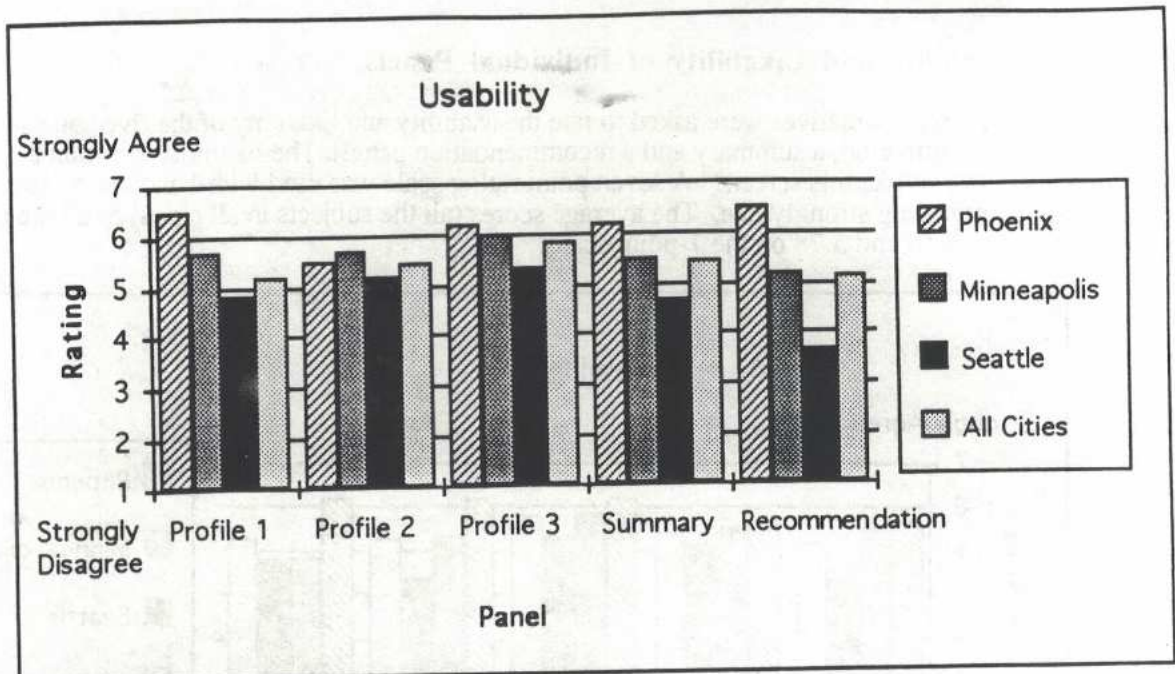


Fig 8. Usability ratings for the panels of TIPS

It is too soon to draw any definite conclusion due to the non-random selection of reps included in the study. However, the following section provides some valuable insights into the areas that offer opportunity for improvement.

4.5 Individual Interview Comments

A variety of profile items and categories, window features, navigational and general systems issues were explored during the individual interviews. The comments were coded, transcribed and unitized from the raw interview data to index cards. The cards were then sorted into comments and suggestions of different categories. Negative comments, together with suggestions to improve the system were separated and reported in these sections so that action items can be developed from the results. The major categories identified were as follows:

- profile items and categories
- visual organization and interface elements
- task compatibility and work flow
- overall system issues ranging from training to the benefits of using TIPS

4.5.1 Profile Items and Categories

The comment of each profile category was grouped and the number of comments was counted.

Profile items with 4 or more negative comments (problems and suggestions to improve) were graphed for better readability. The comments were subdivided into different classes:

- ambiguous - the meaning of the label of the profile item is ambiguous and the rep may misinterpret it without looking at the explanations provided by the help message
- rarely used - the profile item is rarely used by the sales rep. in the course of consulting with their customers
- different category - the profile item belongs to another category
- don't know - the sales rep. do not know the meaning of the label
- inadequate - the profile item is inadequate; expansion is needed
- redundant - the profile item overlaps with another item
- indirect - the profile item can be replaced by a more appropriate item directed to customer's need
- suggestions - suggestions to improve the label of the profile item

The number of comments associated with a profile item will serve as an indicator of the priority of that item to be reworked for better understanding. High priority will be assigned to those items with 6 or more comments, medium priority to 4 and 5 and low to items with comments below 4. In the case that a viable solution is apparent, it will be given in the recommendation column of the table.

The results from the ratings done in Phoenix targeted the following categories for further studies: Business Info, Layout, Concerns, Easy Calling, Usage, Control Calls, Ans Methods, Equipment and Image. The following graph showed in decreasing number of negative comments and suggestions for improvement collected at Phoenix and the subsequent sessions at Minneapolis and Seattle.

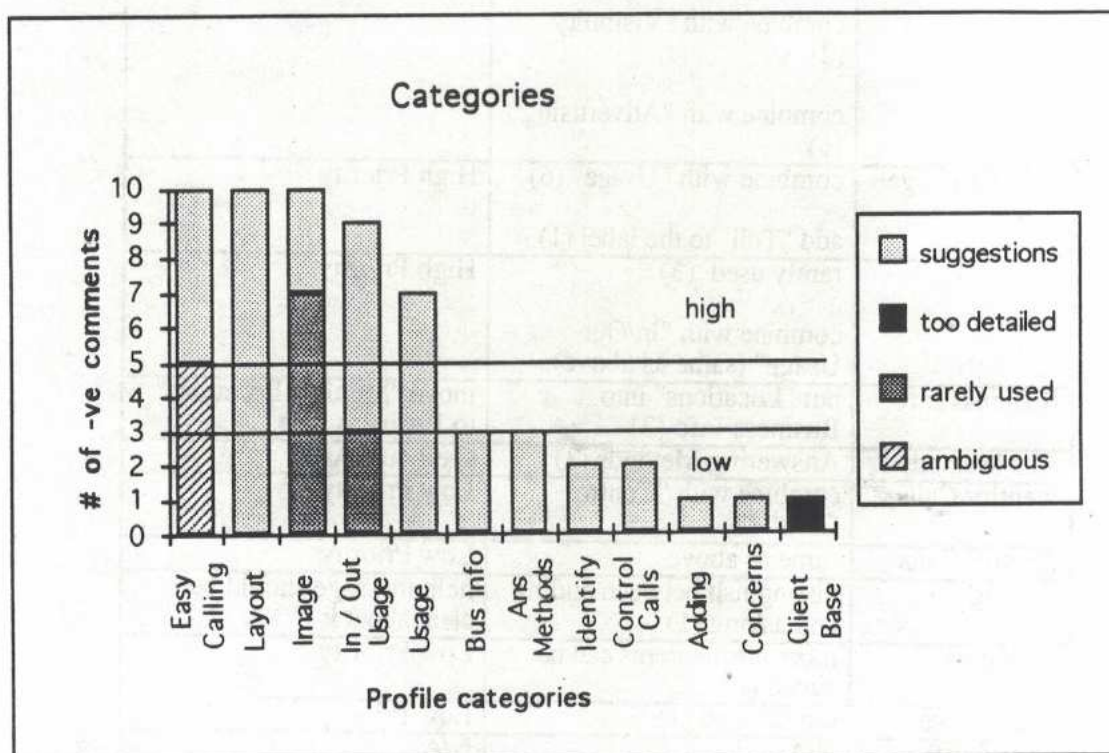


Fig 9. # of suggestions and negative comments vs. profile categories

Categories	Comments & Suggestions	Recommendation
Easy Calling	ambiguous (5) put "Easy Access" and "Easy Recognition" under a new category "Recognition" and "Freq Dialed #", "Message Only", "Conference" and "Transfer" to "Easy Calling" (4) move "Easy Calling" to "Layout" or "Concerns" (1)	High Priority
Layout	prefer "Floor Plans" (6) prefer "Layout" (4)	use both labels or put both in help messages
Image	rarely used (7) combine with "Visibility" (2) combine with "Advertising" (1)	High Priority
In / Out Usage	combine with "Usage" (6) add "Toll" to the label (1)	High Priority
Usage	rarely used (3) combine with "In/Out Usage" (same as above)	High Priority
Business Info	put "Locations" into Business Info (3)	move "Multiple Locations" to Business Info
Ans Methods	Answering Methods (3)	spell out "Ans"
Identify Calls	combine with "Control Calls" (2)	Low Priority
Control Calls	same as above	Low Priority
Adding	distinguish between adding and asking (1)	help message should be clear about it
Concerns	more profile items can be added (1)	Low Priority
Client Base	too detailed (1)	Low Priority
Out of Office	N/A	N/A
Equipment	N/A	N/A
Visibility	N/A	N/A
Advertising	N/A	N/A

Table 11. Suggestions and negative comments of profile category

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4.5.1.2 Business Info

The Phoenix study identified the following items for further investigation: Market Test, Shared Users, Shared Equipment and Work Home Occasionally. Some more items were commented upon in later sessions.

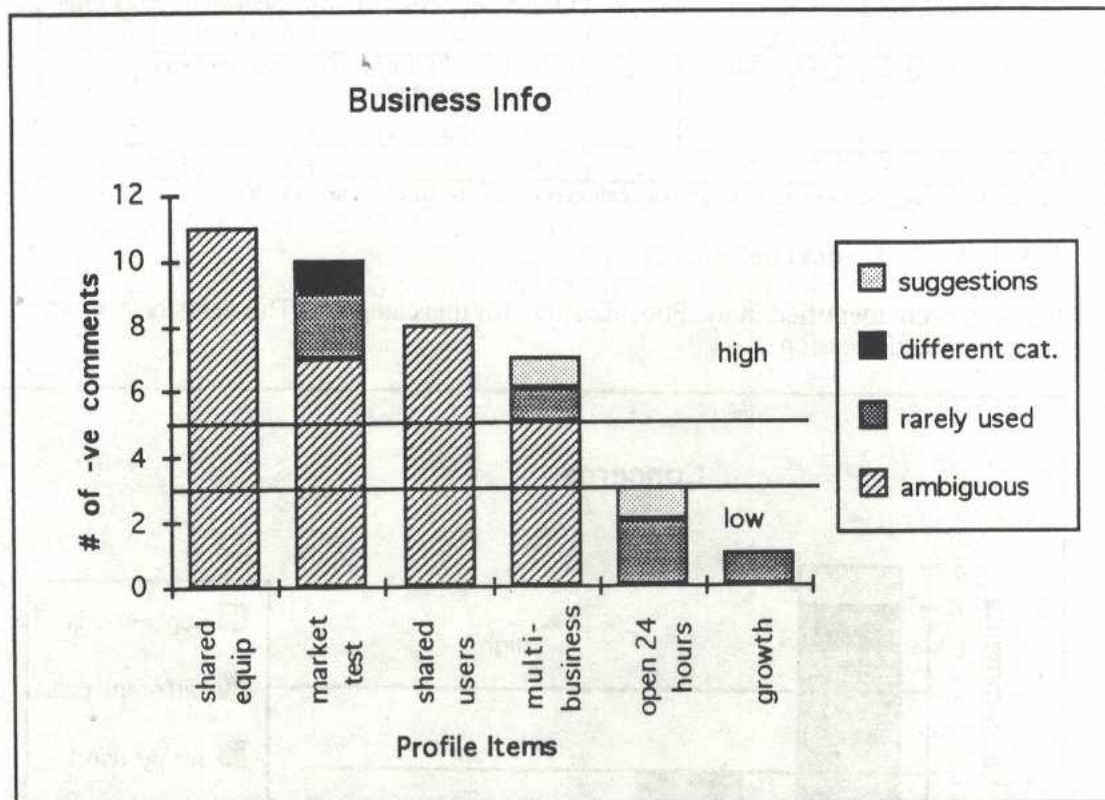


Fig 10. # of suggestions and negative comments vs. profile items in Business Info

The following table summarized the distribution of negative comments and suggestions for improvement on profile items:

Profile Item	Comment	Freq	Suggestions	Recommendation
Shared Equipment	ambiguous	11	N/A	High Priority
Market Test	ambiguous	7	N/A	High Priority
	rarely used	2		
	belong to CONCERN	1		
Shared Users	ambiguous	8	N/A	High Priority

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Multi-business	ambiguous	5	DBAs (1)	High Priority
	rarely used	1		
Locations	rarely used	2	Multiple Locations Other Services (3)	change to "Multiple Locations" or "Other Services"
	ambiguous	1		
Open 24 Hours	rarely used	2	expand to "Hours of Operation". Use Option Menu (1)	Low Priority
Growth	rarely used	1	N/A	Low Priority

Table 12. Suggestions and negative comments of profile items in Business Info

4.5.1.3 Concerns

No items were identified in the Phoenix study for this category. The items below came mostly from the Seattle session.

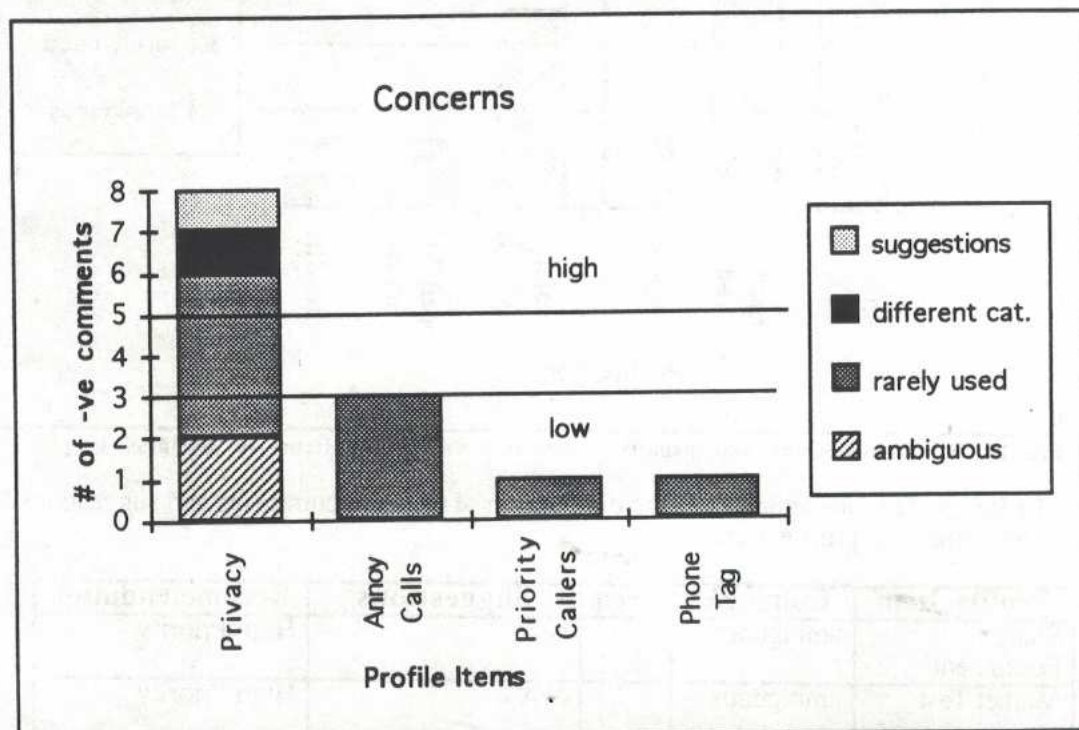


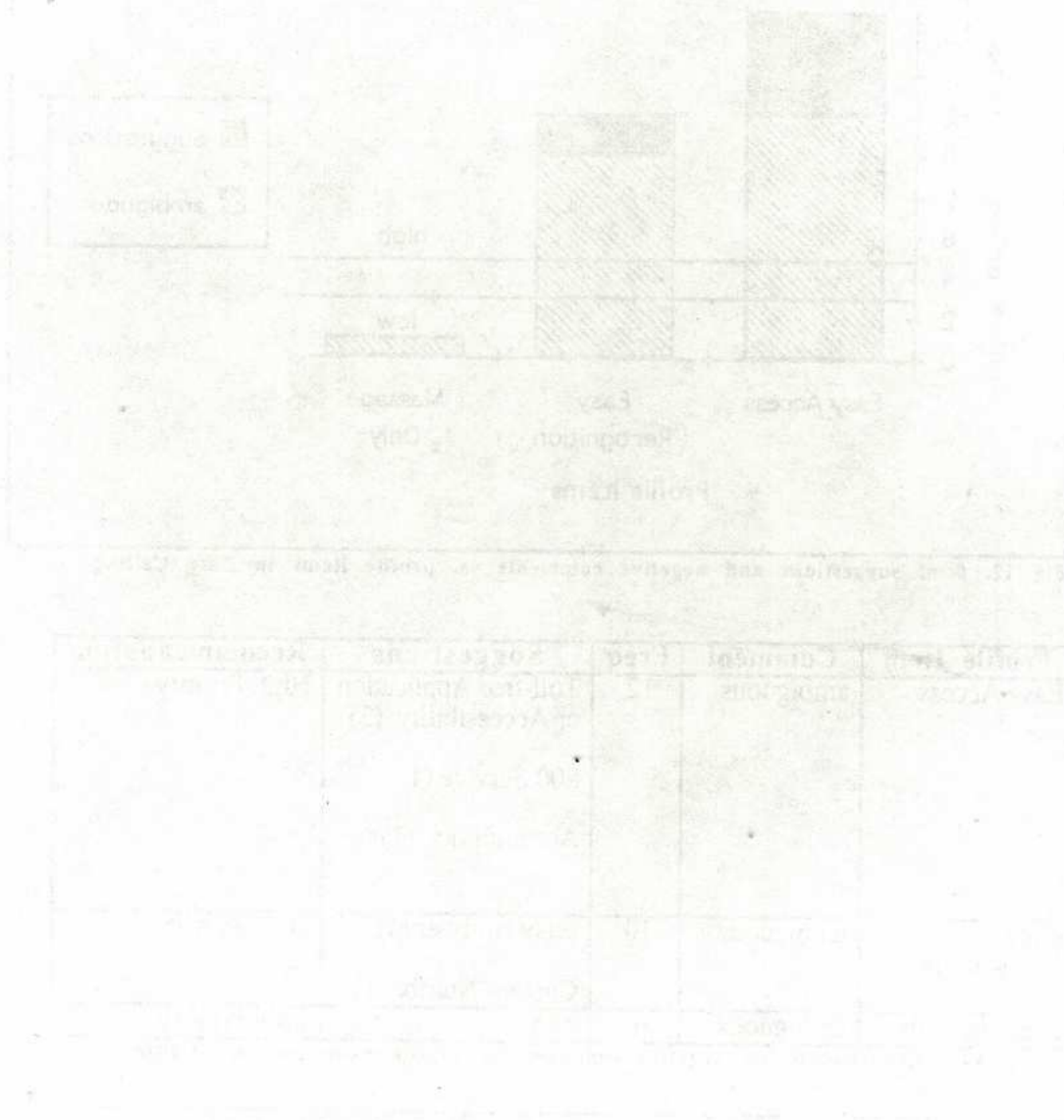
Fig 11. Number of suggestions and negative comments vs. profile items in Concerns

Profile Item	Comment	Freq	Suggestions	Recommendation
Privacy	rarely used	4	Internal vs. External (1)	High Priority
	ambiguous	2		
	related to EQUIPMENT	1		

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Annoyance Calls	rarely used	3	N/A	Low Priority
Priority Callers	rarely used	1	N/A	Low Priority
Phone Tag	rarely used	1	N/A	Low Priority

Table 13. Suggestions and negative comments of profile items in Concerns



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4.5.1.4 Adding

No problematic items were reported.

4.5.1.5 Easy Calling

The Phoenix study identified Easy Access and Easy Recognition for further investigation. Suggestions were also made in Minneapolis that the items under this category can be split.

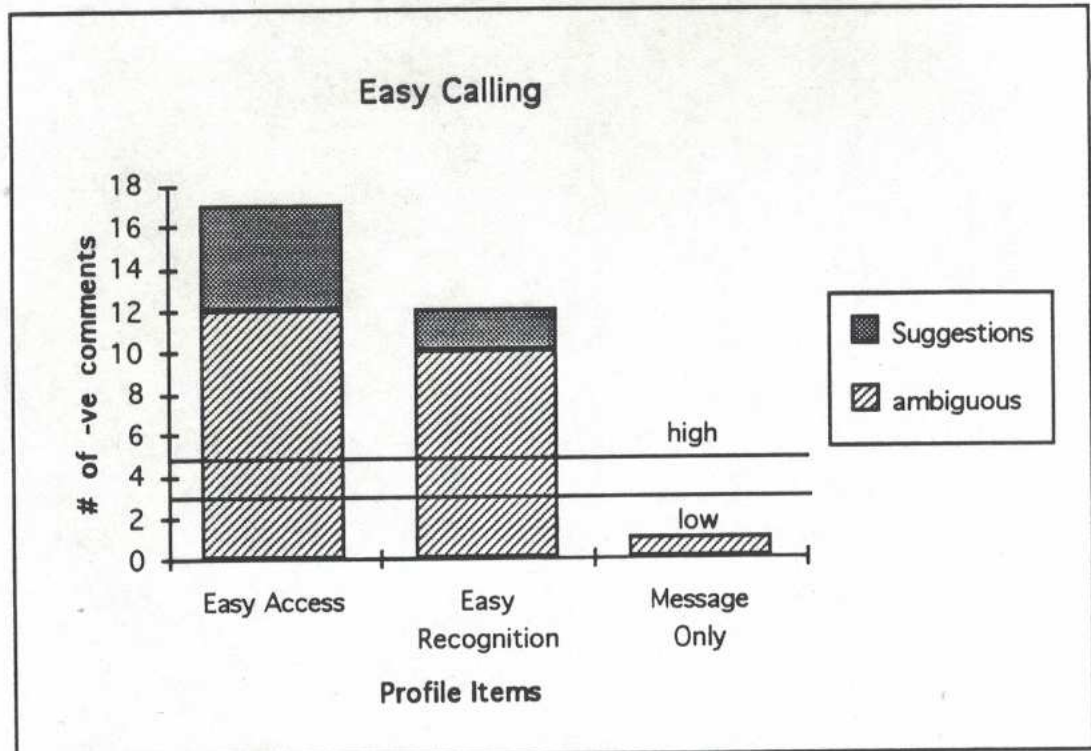


Fig 12. # of suggestions and negative comments vs. profile items in Easy Calling

Profile Item	Comment	Freq	Suggestions	Recommendation
Easy Access	ambiguous	12	Toll-free Application or Accessibility (3) 800 Service (1) Accessibility (1)	High Priority
Easy Recognition	ambiguous	10	Easy Number (1) Custom Number (1)	High Priority
Message Only	ambiguous	1	N/A	Low Priority

Table 14. Suggestions and negative comments of profile items in Easy Calling

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4.5.1.4 Adding

No problematic items were reported.

4.5.1.5 Easy Calling

The Phoenix study identified Easy Access and Easy Recognition for further investigation. Suggestions were also made in Minneapolis that the items under this category can be split.

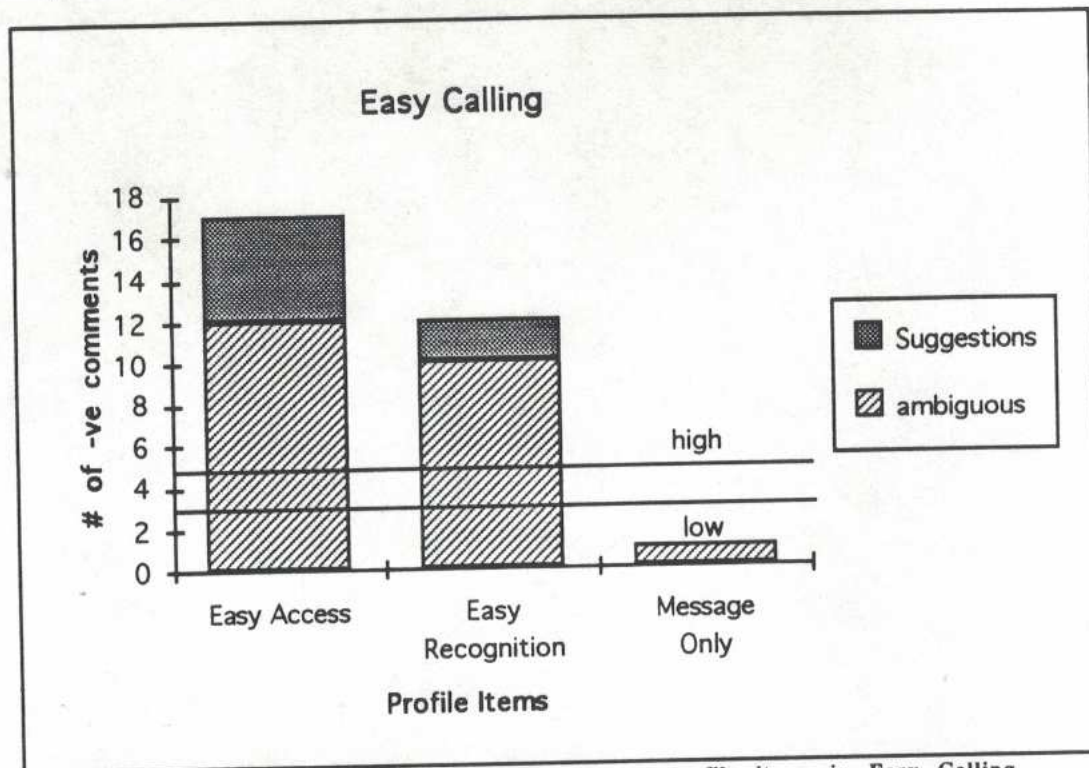


Fig 12. # of suggestions and negative comments vs. profile items in Easy Calling

Profile Item	Comment	Freq	Suggestions	Recommendation
Easy Access	ambiguous	12	Toll-free Application or Accessibility (3) 800 Service (1) Accessibility (1)	High Priority
Easy Recognition	ambiguous	10	Easy Number (1) Custom Number (1)	High Priority
Message Only	ambiguous	1	N/A	Low Priority

Table 14. Suggestions and negative comments of profile items in Easy Calling

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4.5.1.6 In / Out Usage

Data In and Data Out were identified in the Phoenix session.

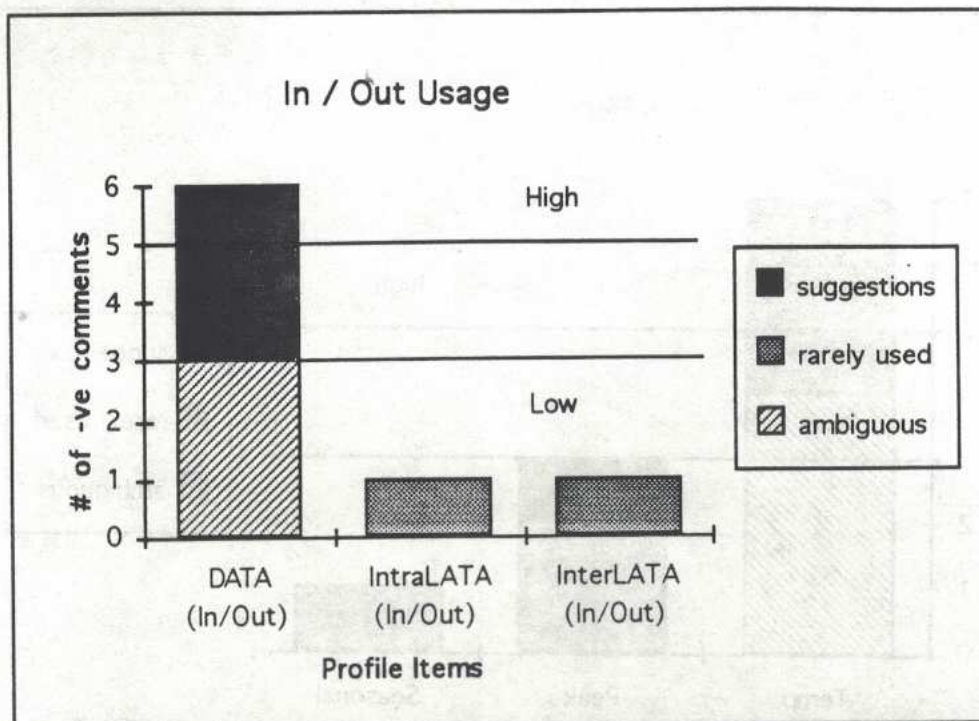


Fig 13. # of suggestions and negative comments vs. profile items in In/Out Usage

Profile Item	Comment	Freq	Suggestions	Recommendation
Data (In/Out)	ambiguous	3	Modem and Fax (2)	High Priority
			Office Machine Use (1)	
IntraLATA	rarely used	1	N/A	Low Priority
InterLATA	rarely used	1	N/A	Low Priority

Table 15. Suggestions and negative comments of profile items in Business Info

4.5.1.7 Usage

Temp Absence and Peak Usage were identified in Phoenix session.

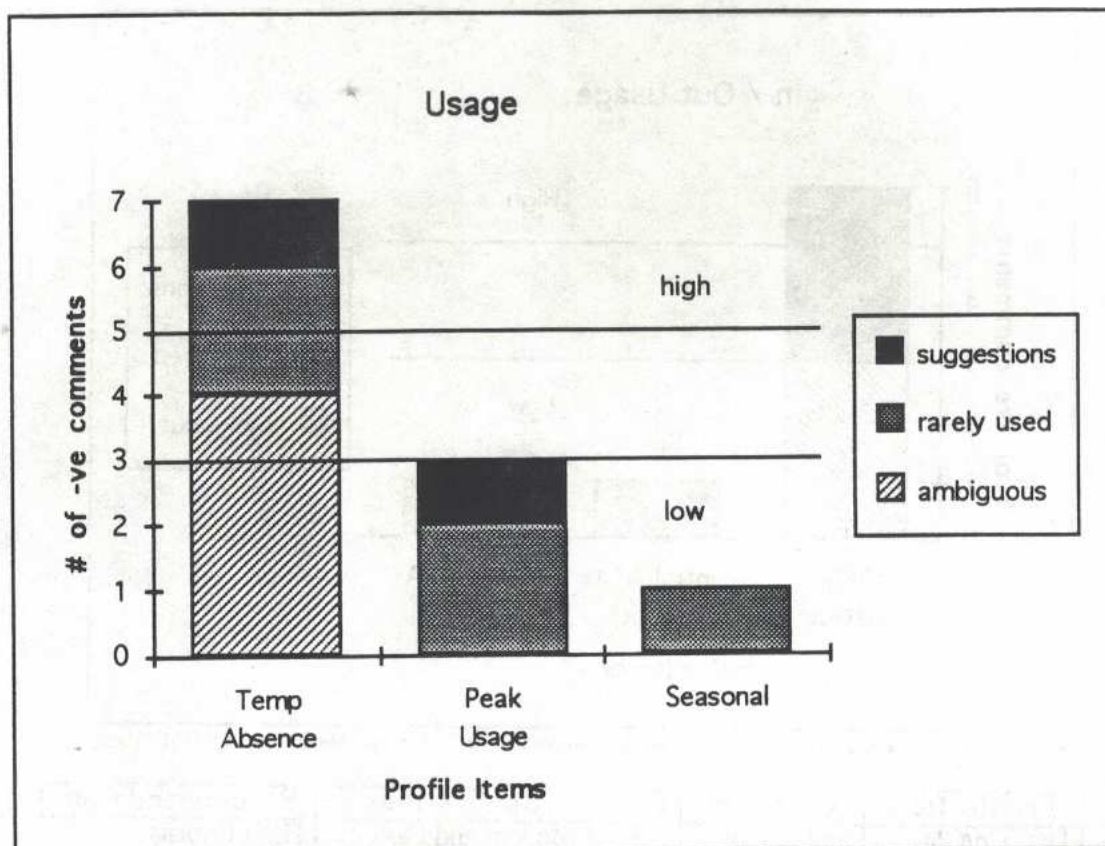


Fig 14. # of suggestions and negative comments vs. profile items in Usage

Profile Item	Comment	Freq	Suggestions	Recommendation
Temp Absence	ambiguos	4	Absence/Coverage or Phone Coverage (1)	High Priority
	rarely used	2		
Peak Usage	rarely used	2	Option Menu -daily, weekly and monthly (1)	Low Priority
Seasonal	rarely used	1	N/A	Low Priority

Table 16. Suggestions and negative comments of profile items in Business Info

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4.5.1.8 Identify Calls

Bus/Personal and Screen were identified in Phoenix study.

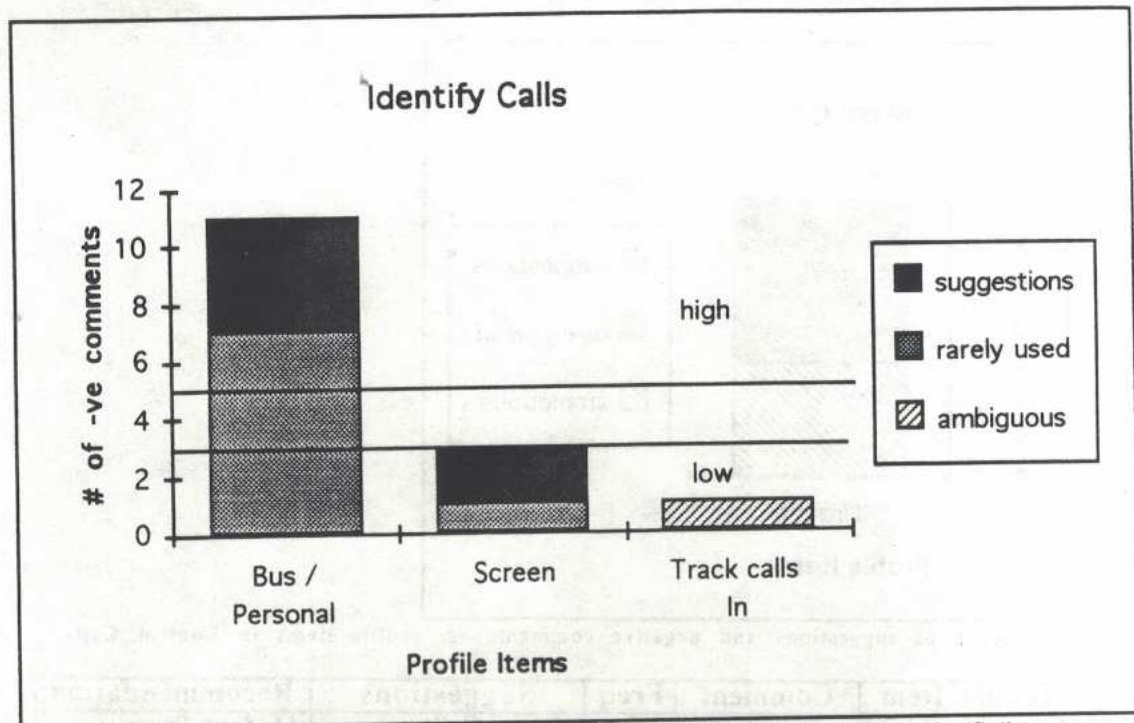


Fig 15. # of suggestions and negative comments vs. profile items in Identify Calls

Profile Item	Comment	Freq	Suggestions	Recommendation
Bus / Personal	rarely used	7	In vs. Out (2)	High Priority
			Owner vs. Employees (1)	
			Distribution of Bus / Personal Calls (1)	
Screen	rarely used	1	In vs. Out (1)	Low Priority
			Home vs. Business (1)	
Track Calls In	ambiguous	1	N/A	Low Priority

Table 17. Suggestions and negative comments of profile items in Identify Calls

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4.5.1.9 Control Calls

Urgent was identified in the Phoenix study.

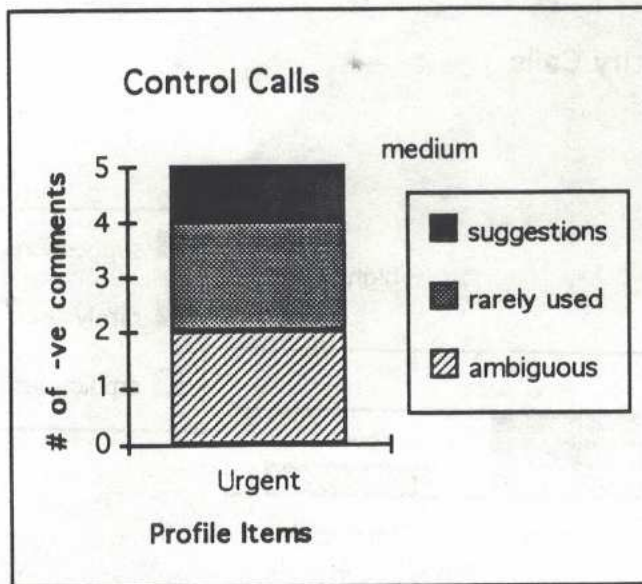


Fig 16. # of suggestions and negative comments vs. profile items in Control Calls

Profile Item	Comment	Freq	Suggestions	Recommendation
Urgent	ambiguous	2	Urgent Calls or Messages, Immediate Responses, Priority Calls (1)	Medium Priority
	rarely used	2		

Table 18. Suggestions and negative comments of profile items in Control Calls

4.5.1.10 Ans Methods

Self, others and After Hours were identified in the Phoenix study.

Profile Item	Comment	Freq	Suggestions	Recommendation
TAD	don't know	1	N/A	include explanation in Help Message
TAS	don't know	1	N/A	include explanation in Help Message
After Hours	remove separator	1	N/A	Low Priority

Table 19. Suggestions and negative comments of profile items in Ans Methods

4.5.1.11 Out of Office

Work out of Office, Voice, Message Notify and On Call were identified in Phoenix.

Profile Item	Comment	Freq	Suggestions	Recommendation
Voice	indirect	1	Assume Voice when Work Out of Office is clicked (1)	Low Priority
	redundant	1		
Data	ambiguous	2	N/A	Low Priority
On Call	ambiguous	2	N/A	Low Priority
Message Notify	redundant	1	Message Notify and On Call are similar (1)	Low Priority

Table 20. Suggestions and negative comments of profile items in Out of Office

4.5.1.12 Equipment

Adjunct was identified in Phoenix study.

Profile Item	Comment	Freq	Suggestions	Recommendation
Adjunct	ambiguous	3	N/A	Low Priority
Cordless Phone	rarely used	2	N/A	Low Priority

Table 21. Suggestions and negative comments of profile items in Equipment

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4.5.1.13 Image

Both Big and Customized were identified in Phoenix study.

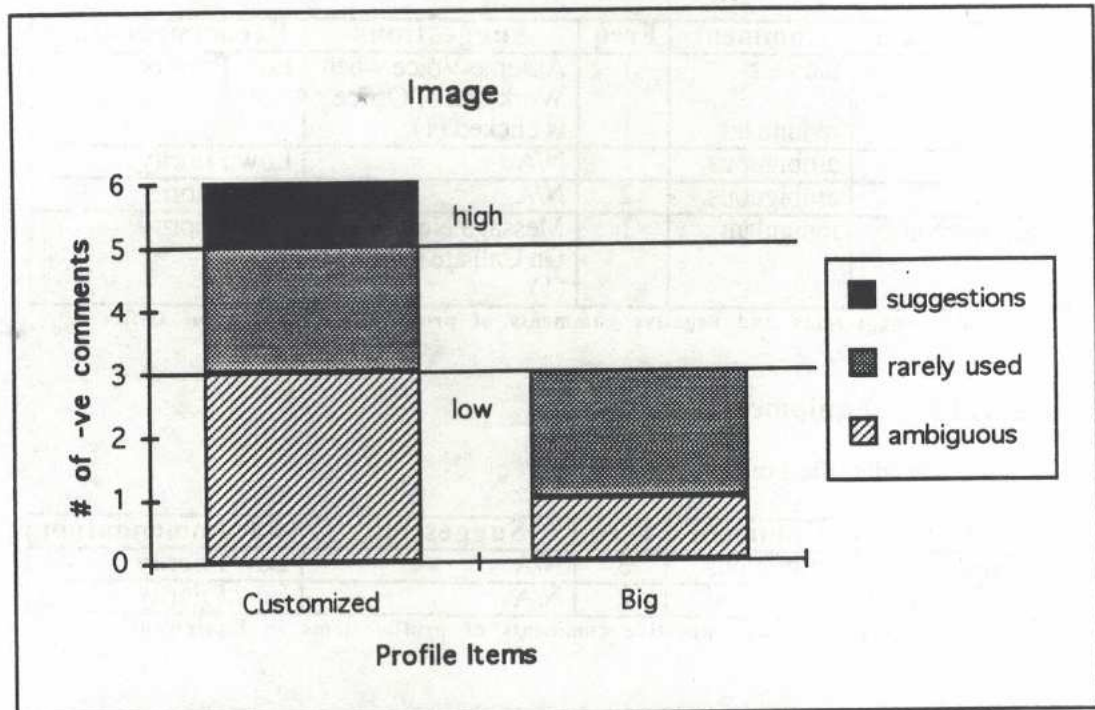


Figure 17. Number of suggestions and negative comments vs. profile items in Image

Profile Item	Comment	Freq	Suggestions	Recommendation
Customized Image	ambiguos	3	Professional Image (1)	High Priority
	rarely used	2		
Big Image	rarely used	2	N/A	Low Priority
	ambiguos	1		

Table 22. Suggestions and negative comments of profile items in Image

4.5.1.14 Visibility

No item was identified in Phoenix for this category.

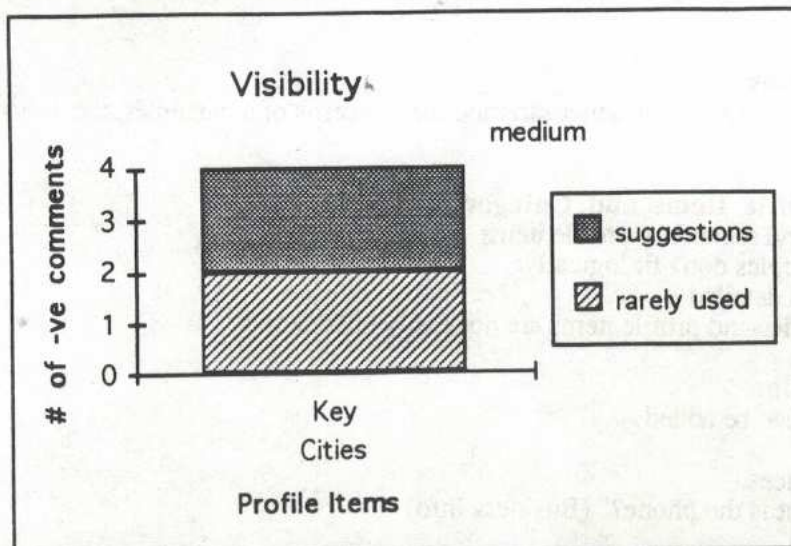


Fig 18. Suggestions and negative comments of profile items in Visibility

Profile Item	Comment	Freq	Suggestions	Recommendation
Key Cities	rarely used	2	Geographic Areas (1) Regional (1)	Medium Priority

Table 23. Suggestions and negative comments of profile items in Visibility

4.5.1.15 Advertising

No item was identified in Phoenix for this category.

Profile Item	Comment	Freq	Suggestions	Recommendation
Multimedia	rarely used	3	N/A	Low Priority

Table 24. Suggestions and negative comments of profile items in Advertising

4.5.1.16 Client Base

Repeat was identified in Phoenix.

Profile Item	Comment	Freq	Suggestions	Recommendation
Repeat	inadequate	1	Option Menu - frequency of repeated service (1)	Low Priority

Table 25. Suggestions and negative comments of profile items in Client Base

4.5.1.17 General Comments and Suggestions

- 15 out of 18 reps. preferred to control the invocation of TIPS so that they can choose to use it as appropriate
- 13 out of 18 reps. prefer to have a 1-1 correspondence between icons and customer site profile panels

Aging of Profile Items

- aging of profile items (e.g. if a product satisfied the concerns of a customer, the concerns should be dropped)

Organization of Profile Items and Categories

- prioritize primary and secondary profile items
- some of the panel topics don't fit logically
- profile items are too detailed
- some of the categories and profile items are not easily understood

Profile Items Addition

- more profile items can be added
- add foreign listing
- add temporary services
- add "How important is the phone?" (Business Info)
- add "billing method"

Profile Items & Categories Layout

- have all profile items on a page
- customization of layout of categories
- box titles of categories

4.6 Summary & Recommendation Panels Comments

Item	Comment	Freq	Suggestions	Recommendation
clicked profile items	too fragmented	1	highlight items that need to be covered (1)	Low Priority
panel boxes	too small	1	N/A	Low Priority

Table 26. Suggestions and negative comments of profile items in Summary

Item	Comment	Freq	Suggestions	Recommendation
pricing information	cannot offer immediate pricing information	2	need to be on recommendation panel (2)	Medium Priority
product recommendation	too generic, cut and dried	1	N/A	Low Priority
product benefits	too scripted	1	N/A	Low Priority
sales pitch	don't like	1	N/A	Low Priority

Table 27. Suggestions and negative comments of profile items in Recommendation

Other suggestions include trends of products and services of different types of business.

4.7 Visual Organization and User Interface Elements Comments

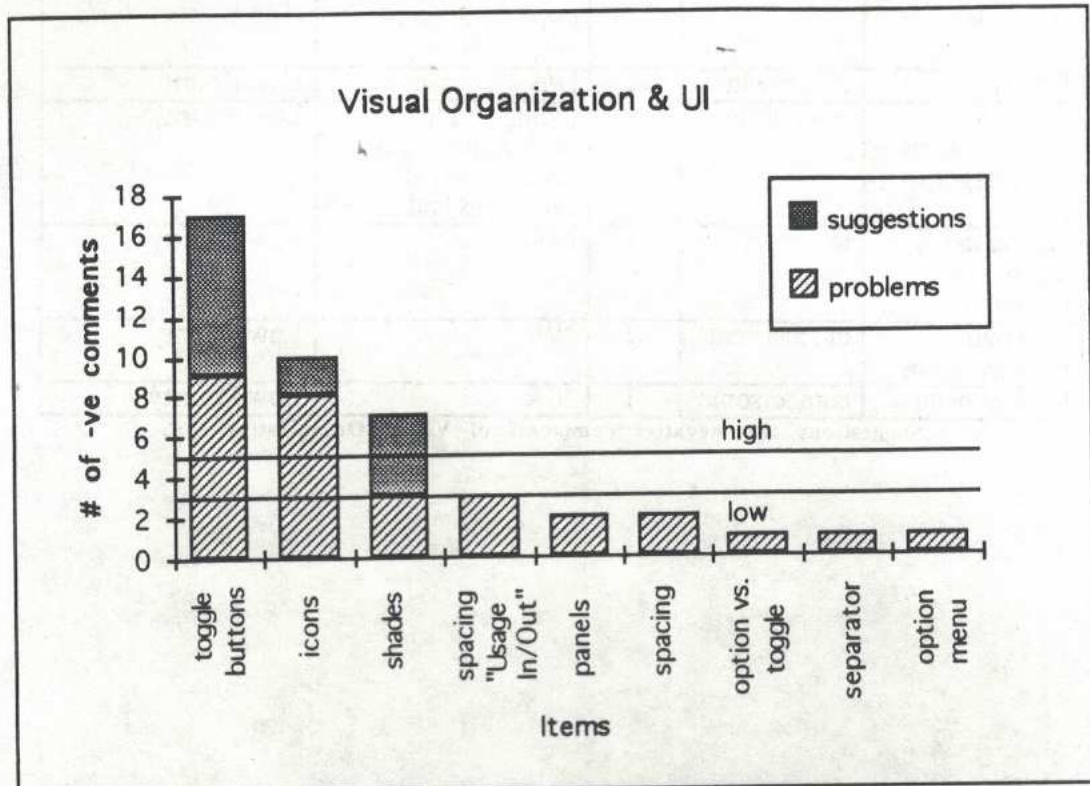


Fig 19. Suggestions and negative comments of Visual Organization

Item	Comment	Freq	Suggestions	Recommendation
Toggle Buttons	hard to see	8	increase contrast (8)	High Priority
	click yes only	1		
Icons	small & busy	2	eliminate some redundant icons and simplify some of those (1) rework the icons (1)	establish 1-1 relationship of icons with panels by dropping 2 icons and redesign some of icons.
	don't like	2		
	lost track	1		
	take time to learn	2		
	don't make sense	1		
Spacing between items in "Usage In/Out"	inadequate	3	(compared to 7 saying spacing is adequate for all categories and items)	Medium Priority

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Shades to differentiate different categories	too close	3	more colors can help (2) customization of colors (2)	Medium Priority
Panels	overwhelm	2	N/A	Low Priority
Spacing between items and categories	inadequate	2	(compared to 7 saying spacing is adequate for all categories and items)	Low Priority
Option menu and toggle buttons	incompatible	1	N/A	Low Priority
Separator between items	inconsistent	1	N/A	Low Priority
Option menu	cumbersome	1	N/A	Low Priority

Table 28. Suggestions and negative comments of Visual Organization

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4.8 Task Compatibility and Work Flow

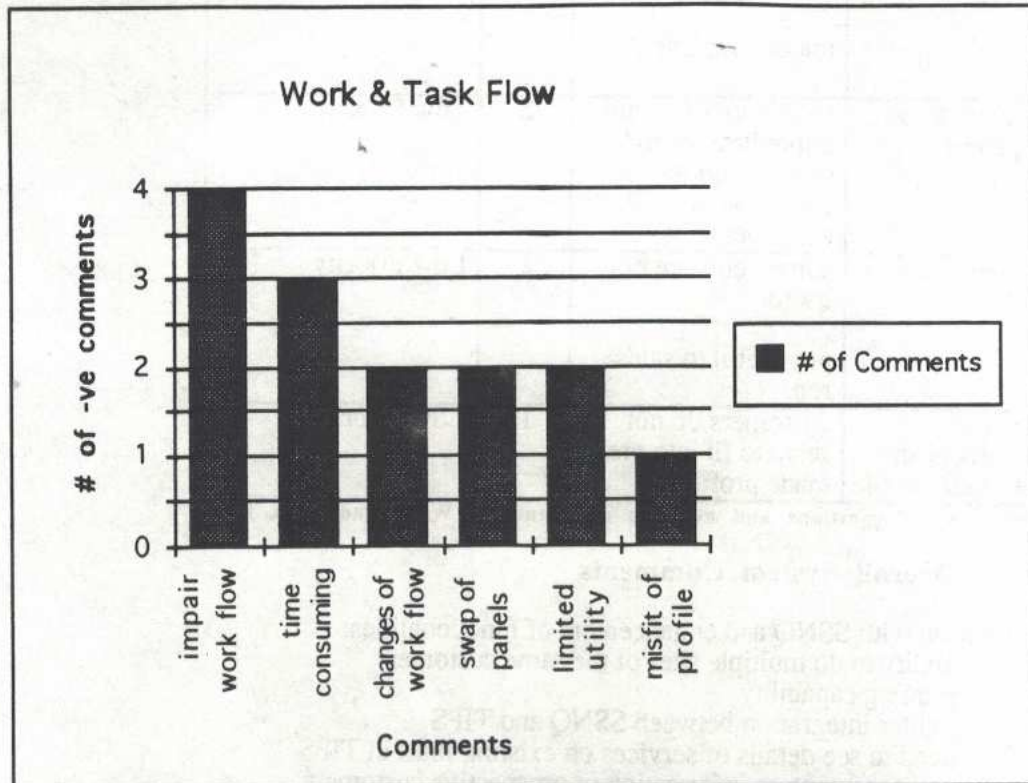


Fig 20. Suggestions and negative comments of Work and Task Flow

Item	Comment	Freq	Recommendation
impair work flow	distracting	4	Medium Priority - redesign TIPS or SSNQ or both to integrate more tightly with product manager
	interfere with my style and flow		
	difficult to work smoothly		
	too many steps		
long time of completion	takes a lot of time	3	Medium Priority
	time consuming to complete		
	customers will not be patient		

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changes of current task	drastic change of what we are currently doing major adjustment	2	Low Priority
frequent swap of panels	profile items hit are unpredictable and depends on the conversation of the customers	2	Low Priority
limited utility of TIPS	some items are not useful not useful to sales rep.	2	Low Priority
misfit of customers and pre-made profile	customers do not seem to fit into pre-made profiles	1	Low Priority

Table 29. Suggestions and negative comments of Work and Task Flow

4.9 Overall System Comments

Integration with SSNQ and enhancement of functionalities:

- ability to do multiple sites of the same customer
- printing capability
- tighter integration between SSNQ and TIPS
- need to see details of services on existing lines in TIPS
- ability to capture information of prospective customers
- need personal or BOSS notes

Help Messages:

- highlight help messages
- help messages can be turned on and off

Training Issues:

- showed and explained more benefits and advantages to the rep.
- introduce panels gradually

Policy Issues:

- office accessibility versus time to consult

4.10 Overall System Value

The following table showed some of the positive comments from the sales reps participating in the study:

Item	Comment	Source
Benefits	wonderful for new products	Min
	help large team concept & new products	Min
	help consultative sales	Phx, Min
	allow me to establish a profile of my customer and remember products based on their needs and desires	Phx, Min
	made our job a little easier	Phx
	product benefits are helpful	Min
Training / New People	good for new people and training	Min, Phx
	good for rep. who are not comfortable with sales	
Design & Idea	got down to the sales rep. language	Phx
	cannot believe how well it is put together, excellent job!	Phx
	actually appears to be very user friendly and I don't need a computer science degree to help me use the system	Min
	wonderful concept	Phx
use of mouse	extremely simple and reduce carpal-tunnel syndrome	Phx

Table 30. Positive comments of TIPS

4.11 Other Suggestions from CCOE Observers

The following suggestions were made by two CCOE observers:

Profile Panel Organization

First Panel Business Info
Categories Business Info, Layout, Ans Methods, Out of Office

Second Panel Business Needs
Categories Adding, In/Out Usage, Usage, Identify Calls, Control Calls, Equipment

Third Panel Customer Needs
Categories Concerns, Easy Calling, Image, Visibility, Advertising and Client Base

Profile Item and categories Labels

Problems Multi-Business, Shared Equipment, Easy Calling, IntraLATA In/Out, InterLATA In/Out, Usage, Urgent, Temp Absence
Suggestions replace Urgent by prioritize calls or urgent control
 expand Screen to Screen Calls
 replace Temp Absence by Temp Closure
 expand Usage to Usage Patterns
 expand Image to Company Image

5.0 General Recommendations and Conclusions

Based on the results obtained from the usability studies and the current state of TIPS, the following recommendations can be highly beneficial to the project:

- Rework the problematic profile items and categories identified as high and medium priority. This will involve several participatory design sessions with the SMEs and the knowledge engineers.
- Identify and drop profile items that play a minimal role in consultative sales activities (e.g., seldom used by the rep as memory support or building rapport) and not necessary for the knowledge-based engine. It will shorten the list of profile items and reduce the possible impact on work flow and contact time.
- Additional work on refining the names (labels) of needs (capabilities) used on recommendation window and integrating the recommendation window with SSNQ product manager.
- The above work can be performed with the help of subject matter experts as well as sales reps from business office for informal validation. It is recommended that several sales reps (6) from Denver Service Center can participate in focus group design and validation sessions once a week (for a period of about a month) to iterate on the work done above.
- After the changes are implemented on the TIPS Prototype, usability studies should be conducted in various cities to elicit further feedback from different reps. It is also recommended that the sales reps can have real hands-on experience with the prototype through several scenarios so that more focused feedback on TIPS can emerge.
- The Home Office Channel of SBG will be a fertile source to study the usability and effectiveness of CASH (the "TIPS" variant for the channel). The sales consultants of the channel will be using CASH to take live calls in March. The products sold and the products recommended will also be tracked. An intensive usability study on CASH will be very helpful to TIPS after the consultants learn and use the system.

Appendix A: Interview Questions

TIPS Focused Usability Questions (Iteration from Phoenix Studies)

1. Do you prefer TIPS Manager pop up automatically for you whenever you select a customer site?

2. Is the use of two icons to access one panel (like the business info and concerns panel) confusing or helpful to you?

3. On the First Profile Panel, what do you think about the names and descriptions of the following items:

a. Market Test

b. Shared Users

c. Shared Equipment

d. Work Home Occas

e. Locations

f. Easy Access

g. Easy Recognition

4. Will you classify the items under "Business Info" in another way?

5. What do you think about the labels "Layout" or "Floor Plans"? Which one will you prefer to use? Any other suggestions?

6. Will you classify the items under "Easy Calling" in another way?

7. Can you think of any alternatives to the label and description of "Easy Calling" and "Business Info"?

8. What is your general impressions of this panel?

9. On the Second Profile Panel, what do you think about the names and descriptions of the following items:

a. Peak Usage

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b. Temp Absence

c. Bus / Personal

d. Screen

e. Urgent

f. Self

g. Others

h. After Hours

10. Will you classify the items under "Usage" in another way?

11. Will you classify the items under "Ans Methods" in another way?

12. What is your impression of this panel?

13. On the Third Profile Panel, what do you think about the names and descriptions of the following items:

a. Adjunct

b. Big (Image)

c. Customized (Image)

14. Will you classify the items under "Image" in another way?

15. What is your impression of this panel?

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TIPS Usability Questions (Iterations from Phoenix & Minneapolis Studies)

1. Do you prefer TIPS Manager pop up automatically for you whenever you select a customer site?

2. Is the use of two icons to access one panel (like the business info and concerns panel) confusing or helpful to you?

3a. Please spend some time looking at the labels of the profile items panels and comment on any items that are confusing or not self-explanatory to you. How can they be improved?

Panel 1

Panel 2

Panel 3

3. On the First Profile Panel, what do you think about the names and descriptions of the following items:

Multi-Business

Market Test

Growth

Shared Users

Shared Equipment

Work Home Occas

Locations

Phone Tag

Privacy

Adding Jacks

Easy Access

Easy Recognition

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Message Only

Transfer

4. Will you classify the items under "Business Info" in another way?

5. What do you think about the labels "Layout" or "Floor Plans"? Which one will you prefer to use? Any other suggestions?

6. Will you classify the items under "Easy Calling" in another way?

6a. A suggestion made in earlier session : classify "easy access" (labels subject to change) and "easy recognition" under a category called "recognition" (for the clients of customers) and the remaining items under "easy calling" (to make reps. sell easier). What do you feel about this suggestion?

7. Can you think of any alternatives to the label and description of "Easy Calling" and "Business Info"?

8. What is your general impressions of this panel?

9. On the Second Profile Panel, what do you think about the names and descriptions of the following items:
Data In (Out)

Peak Usage

Temp Absence

Bus / Personal

Screen

Urgent

Self (Ans Methods)

Others

After Hours

Voice (Out of Office)

Data

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Feb 15, 1994

38

On Call

10. Will you classify the items under "Usage" in another way?

10a. Will you combine "Usage" and "In/Out Usage"?

11. Will you classify the items under "Ans Methods" in another way?

12. What is your impression of this panel?

13. On the Third Profile Panel, what do you think about the names and descriptions of the following items:

Cordless

Adjunct

Big (Image)

Customized (Image)

Key Cities Visibility

Multimedia (Advertising)

14. Will you classify the items under "Image" in another way?

14a. Will you use "Image" when you talk to your customers?

15. What is your impression of this panel?

16. Do you think that the state of the toggle button (whether it is on/off) is easy to spot? Any suggestions?

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39

Appendix B: Rating Forms (Usability Inventory)

TIPS/SSNQ Usability Study &
Presentation Model Development
Conceptual Walkthrough Feedback Form
January, 1994

City : _____
Date : _____

Name (optional) : _____
Years of experience with U S WEST : _____
Years of experience as SBG sales rep : _____

The following questions ask for feedback concerning specific impressions of TIPS. Please give us your honest opinion. Negative feedback is just as important and helpful as positive feedback.

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SSNQ / TIPS Usability Evaluation
Profile Items Panels (Panel 1)

The categories and profile items on the panel are easy to read.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6
					7

Comments: _____

The categories and profile items appear to be organized logically.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6
					7

Comments: _____

It is difficult to find the required profile item or category.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6
					7

Comments: _____

The different color shades for the categories help me to locate the appropriate category.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6
					7

Comments: _____

The presentation of the categories and the profile items facilitates the flow of consultative sales.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6
					7

Comments: _____

This panel will prompt me to ask more questions that facilitate consultative sales.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6
					7

Comments: _____

The panel is ill-organized from my point of view.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6
					7

Comments: _____

The icon representing "Business Info" makes sense to me.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6
					7

Comments: _____

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The icon representing "Concerns" makes sense to me.

Strongly Disagree 1 2 3 4 5 Strongly Agree 6 7

Comments: _____

How much did you like this panel?

Strongly Dislike 1 2 3 4 5 6 Strongly Like 7

Comments: _____

I think this panel would be easy to use.

Strongly Disagree Strongly Agree

1 2 3 4 5 6 7

Comments: _____

Overall Comments:

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**SSNQ / TIPS Usability Evaluation
Profile Items Panels (Panel 2)**

The categories and profile items on the panel are easy to read.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The categories and profile items appear to be organized logically.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

It is difficult to find the required profile item or category.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The different color shades for the categories help me to locate the appropriate category.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The presentation of the categories and the profile items facilitates the flow of consultative sales.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

This panel will prompt me to ask more questions that facilitate consultative sales.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The panel is ill-organized from my point of view.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The icon representing "Line Usage" makes sense to me.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

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is panel?

Strongly Like

1

2

3

4

5

6

1

Comments: _____

•

Strongly Agree

1

2

3

4

5

6

11

Comments: _____

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**SSNQ / TIPS Usability Evaluation
Profile Items Panels (Panel 3)**

The categories and profile items on the panel are easy to read.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The categories and profile items appear to be organized logically.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

It is difficult to find the required profile item or category.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The different color shades for the categories help me to locate the appropriate category.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The presentation of the categories and the profile items facilitates the flow of consultative sales.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

This panel will prompt me to ask more questions that facilitate consultative sales.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The panel is ill-organized from my point of view.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The icon representing "Equipment" makes sense to me.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

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1

2

3

4

5

6

6

1

.....

panel2

1

2

3

4

2

6

1

114

I think this panel would be easy to use.

easy to use.

1

2

3

4

2

6

2

21

Overall Comments:

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**SSNQ / TIPS Usability Evaluation
Summary Panel**

The categories and profile items on the panel are easy to read.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The summary panel is very important to me for a bird's eye view of the profile information of my customers when I present the recommendations.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The presentation of the categories and the profile items facilitates the flow of consultative sales.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

It is difficult to find the required profile item or category.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

The icon representing "Summary" makes sense to me.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

How much did you like this panel?

Strongly Dislike					Strongly Like
1	2	3	4	5	6 7

Comments: _____

I think this panel would be easy to use.

Strongly Disagree					Strongly Agree
1	2	3	4	5	6 7

Comments: _____

Overall Comments

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**SSNQ / TIPS Usability Evaluation
Recommendation Panel**

The recommended products and their functions are easy to read.

Strongly Disagree						Strongly Agree
1	2	3	4	5	6	7

Comments: _____

I find it difficult to understand and use the organizational layout of the panel.

Strongly Disagree						Strongly Agree
1	2	3	4	5	6	7

Comments: _____

The recommended products are logically related to the needs and profile of the customer (site).

Strongly Disagree						Strongly Agree
1	2	3	4	5	6	7

Comments: _____

I find it easy to come up with the appropriate product recommendations for my customer with this panel.

Strongly Disagree						Strongly Agree
1	2	3	4	5	6	7

Comments: _____

The product recommendations and related support information are compatible with the flow of consultation with my customers.

Strongly Disagree						Strongly Agree
1	2	3	4	5	6	7

Comments: _____

This panel will help me to consult and negotiate products effectively with my customers.

Strongly Disagree						Strongly Agree
1	2	3	4	5	6	7

Comments: _____

It is easy to click on the recommended product (or its function) to obtain product benefits (or needs / profile relationship) with the use of a mouse.

Strongly Disagree						Strongly Agree
1	2	3	4	5	6	7

Comments: _____

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The panel will prevent me from conducting consultative sales effectively.

Strongly Disagree 1 2 3 4 5 6 Strongly Agree 7

Comments: _____

The icon representing "Recommendation" makes sense to me.

Strongly Disagree 1 2 3 4 5 6 Strongly Agree 7

Comments: _____

How much did you like this panel?

Strongly Dislike 1 2 3 4 5 6 Strongly Like 7

Comments: _____

I think this panel would be easy to use.

Strongly Disagree 1 2 3 4 5 6 Strongly Agree 7

Comments: _____

Overall Comments

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Overall TIPS Usability Evaluation

I can easily follow the steps of using TIPS to assist consultative sales contact with my customers.

Strongly Disagree Strongly Agree

1 2 3 4 5 6 7

Comments: _____

The panels are organized naturally and compatible with the flow of consultative sales contact.

Strongly Disagree Strongly Agree

1 2 3 4 5 6 7

Comments: _____

The use of TIPS will not facilitate consultative sales contact with my customers.

Strongly Disagree Strongly Agree

1 2 3 4 5 6 7

Comments: _____

I will use TIPS to support my consultative sales contact with my customers.

Strongly Disagree Strongly Agree

1 2 3 4 5 6 7

Comments: _____

Overall speaking, TIPS adds more value to the consultative sales.

Strongly Disagree Strongly Agree

1 2 3 4 5 6 7

Comments: _____

Overall Comments:

Business Solution

Reminder	Manual Forwarding	Interrupt
Calling Cards	CF Variable	Call Waiting
Wire Maintenance	CF Programmable	Add Line + CF BLDA
Time & Materials	Selective CF	

Needs / Profile

need Manual Forwarding because
Concerns Busy Line (Yes)
Out of Office Work Out of Office (Yes)

OK

Help

File Edit View		Help	
          		1475 Lawrence	
Summary			
BUSINESS INFO Growth (Expand) Work at Home Occas		EQUIPMENT Cellular / Mobile (Own) Fax (Own) PC Modem (Own) Ans Mchn (Own)	
CONCERNS Busy Line		ADVERTISING Multimedia	
		CLIENT BASE New Repeat	

File Edit View		Help	
1475 Lawrence			
Equipment & Image			
EQUIPMENT		ADVERTISING	
Single Line Set	Unknown	Yellow Pages	Yes No
Multi Line Set	Unknown	White Pages	Yes No
Cordless Set	Unknown	Multimedia	Yes No
Cellular / Mobile	Unknown	CUSTOMER	Yes No
Pager	Unknown	New	Yes No
Fax	Unknown	Repeat	Yes No
PC Modem	Unknown	Walk-Ins	Yes No
Credit Card Mchn	Unknown	Referrals	Yes No
Ans Mchn	Unknown		
Adjunct	Unknown		
IMAGE		VISIBILITY	
Big	Yes No	Local	Yes No
Customized	Yes No	Key Cities	Yes No
		State	Yes No
		National	Yes No

File Edit View		Help	
1475 Lawrence			
Line Usage			
IN / OUT USAGE		IDENTIFY CALLS	
Local In	Unknown	Bus / Personal	Yes No
Local Out	Unknown	Screen	Yes No
IntraLATA In	Unknown	Track Calls In	Yes No
IntraLATA Out	Unknown	Track Calls Out	Yes No
InterLATA In	Unknown	CONTROL CALLS	
InterLATA Out	Unknown	Restrict Calls In	Yes No
Data In	Unknown	Restrict Calls Out	Yes No
Data Out	Unknown	Urgent	Yes No
USAGE		OUT OF OFFICE	
Seasonal	Yes No	Work Out of Office	Yes No
Peak Usage	Yes No	Voice	Yes No
Low Absence	Yes No	Data	Yes No
		Message Notify	Yes No
		On Call	Yes No



14/5 Lawrence

Business Info & Concerns

BUSINESS INFO	
New / Startup	☐ Yes ☒ No
Moving	☐ Yes ☒ No
Multi Business	☐ Yes ☒ No
Growth	☐ Yes ☒ No
Market Test	☐ Yes ☒ No
Open 24 Hours	☐ Yes ☒ No
Shared Users	☐ Yes ☒ No
Shared Equipment	☐ Yes ☒ No
Work Home Occnly	☐ Yes ☒ No
SBG PROMO	☐ Yes ☒ No
Product Promo	☐ Yes ☒ No

LAYOUT	
Staff #	☐ Yes ☒ No
Depts / Offices	☐ Yes ☒ No
Work Areas	☐ Yes ☒ No
Locations	☐ Yes ☒ No

CONCERNS	
Missed Calls	☐ Yes ☒ No
Busy Line	☐ Yes ☒ No
Phone Tag	☐ Yes ☒ No
Lost Messages	☐ Yes ☒ No
Priority Callers	☐ Yes ☒ No
Privacy	☐ Yes ☒ No
Annoyance Calls	☐ Yes ☒ No

EASY CALLING	
Easy Access	☐ Yes ☒ No
Easy Recognition	☐ Yes ☒ No
Freq Dialed #	☐ Yes ☒ No
Conference	☐ Yes ☒ No
Message Only	☐ Yes ☒ No
Transfer	☐ Yes ☒ No

ADDING	
Line	☐ Yes ☒ No
Equipment	☐ Yes ☒ No
Features	☐ Yes ☒ No
Jacks	☐ Yes ☒ No

Appendix D: TIPS Panels

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